DEV CON VIII INDIA

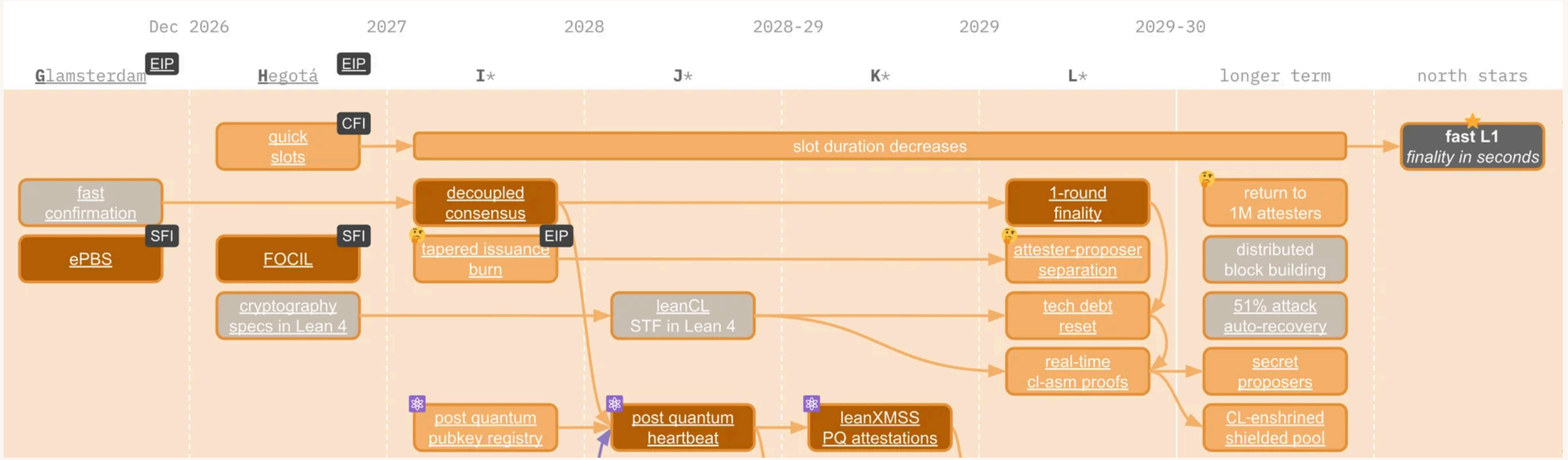
From 16 Minutes to Seconds



A Roadmap to Fast Finality on Ethereum

Yann Vonlanthen · Research Scientist, Ethereum Foundation

The north star: seconds.



strawmap.org

Rebuilding proof of stake, in stages.

16 min
today



1

Decoupled consensus
~4 min

2

One-round finality
< 1 min

3

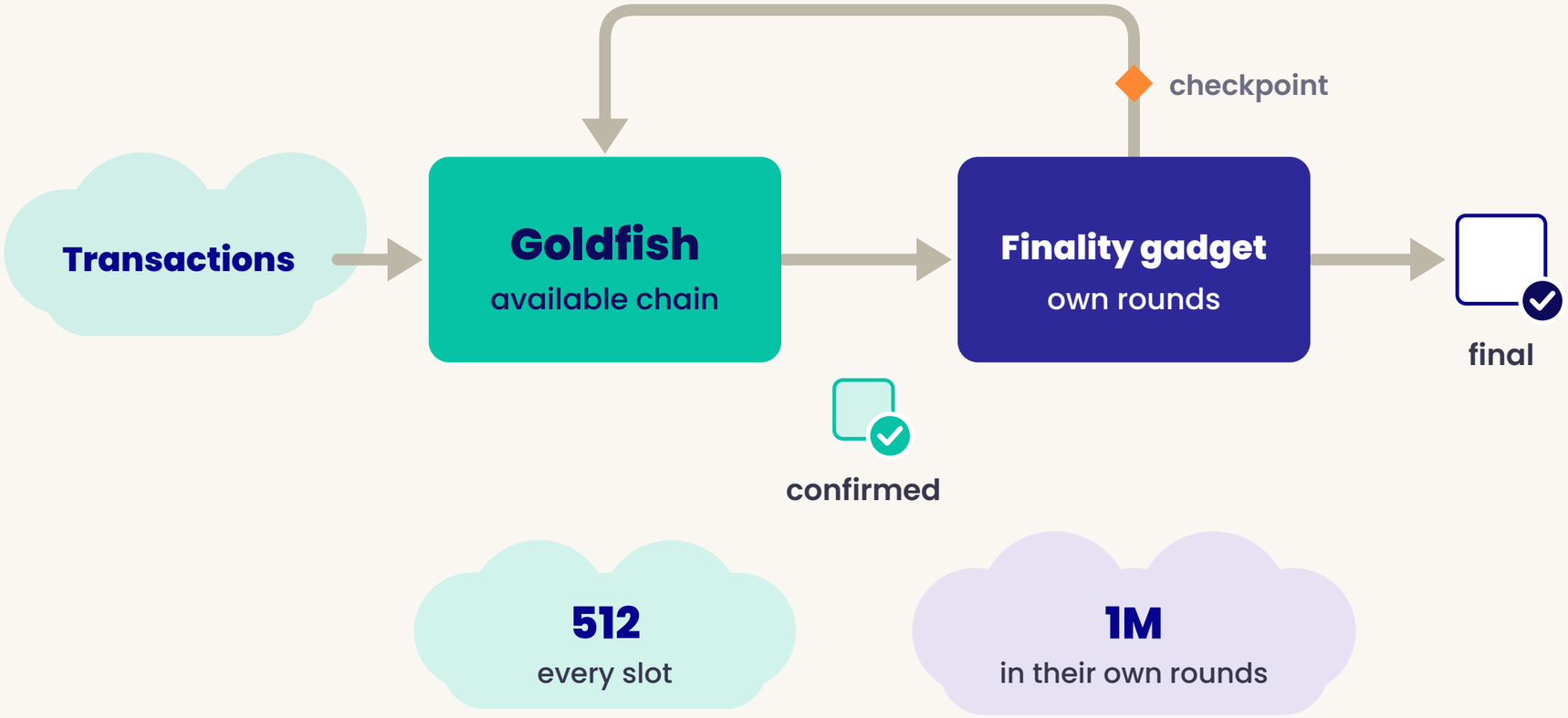
Shorter rounds
seconds

Finality in seconds

STAGE 1 · DECOUPLED CONSENSUS

Stage 1: decouple them

Goldfish runs the chain. Finality gets its own rounds.



STAGE 1 · DECOUPLED CONSENSUS

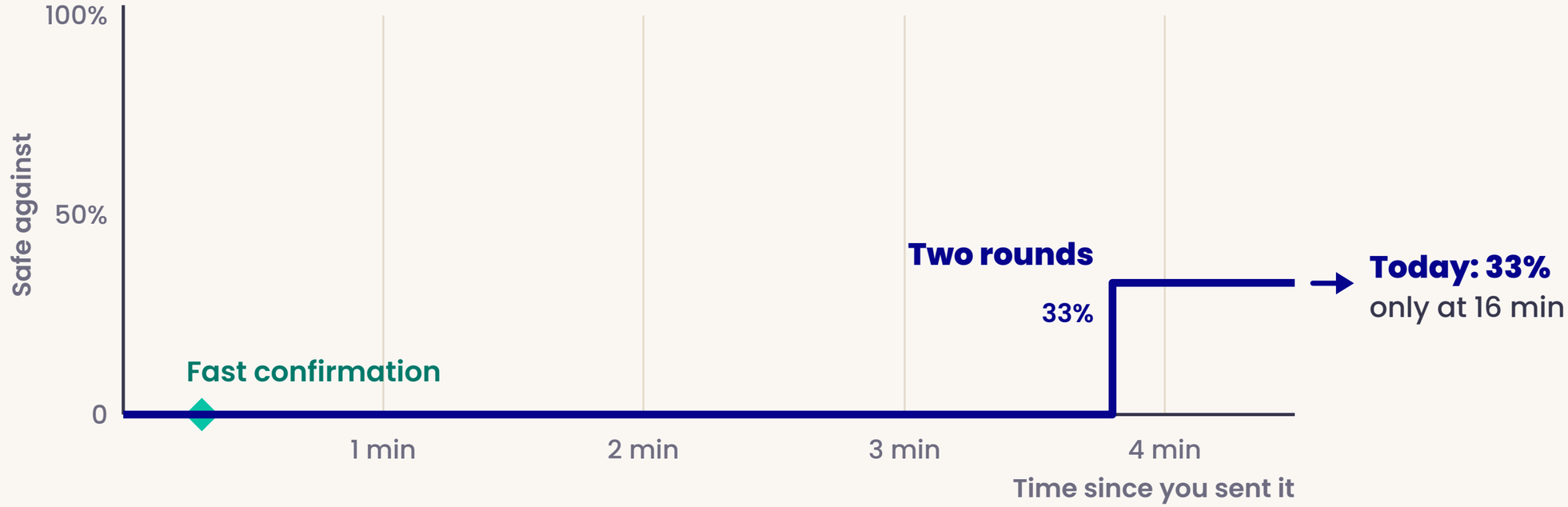
How safe, how soon

final justified fast confirmed new checkpoint

Decoupled consensus
two rounds



3m 48s



EIP-8453

EIPS/eip-8453.md

eip	8453
title	Decoupled Consensus
description	Separate available-chain voting from finality voting to improve reorg resilience and enable faster finality.
author	Francesco D'Amato (@fradamt), Roberto Saltini (@saltiniroberto), Yann Vonlanthen (@yannvon), Mikhail Kalinin (@mkalinin), Ben Edgington (@benjaminion)
discussions-to	https://ethereum-magicians.org/t/eip-8453-decoupled-consensus/30045
status	Draft
type	Standards Track
category	Core
created	2026-10-21

Abstract

This EIP replaces the beacon chain's coupled fork-choice and finality voting with an available-chain protocol and a separate finality gadget. Validators cast head votes and finality attestations as separate duties. The available chain uses recent head votes to select a head and confirm blocks, while locks protect confirmations when votes are missing. The finality gadget justifies and finalizes blocks at quorum-driven logical heights, with attestations sent on a separate round schedule.

And we can be more ambitious.

PQ read

ilability

50% faul

uick slots

one-round finality

STAGE 2 · ONE-ROUND FINALITY

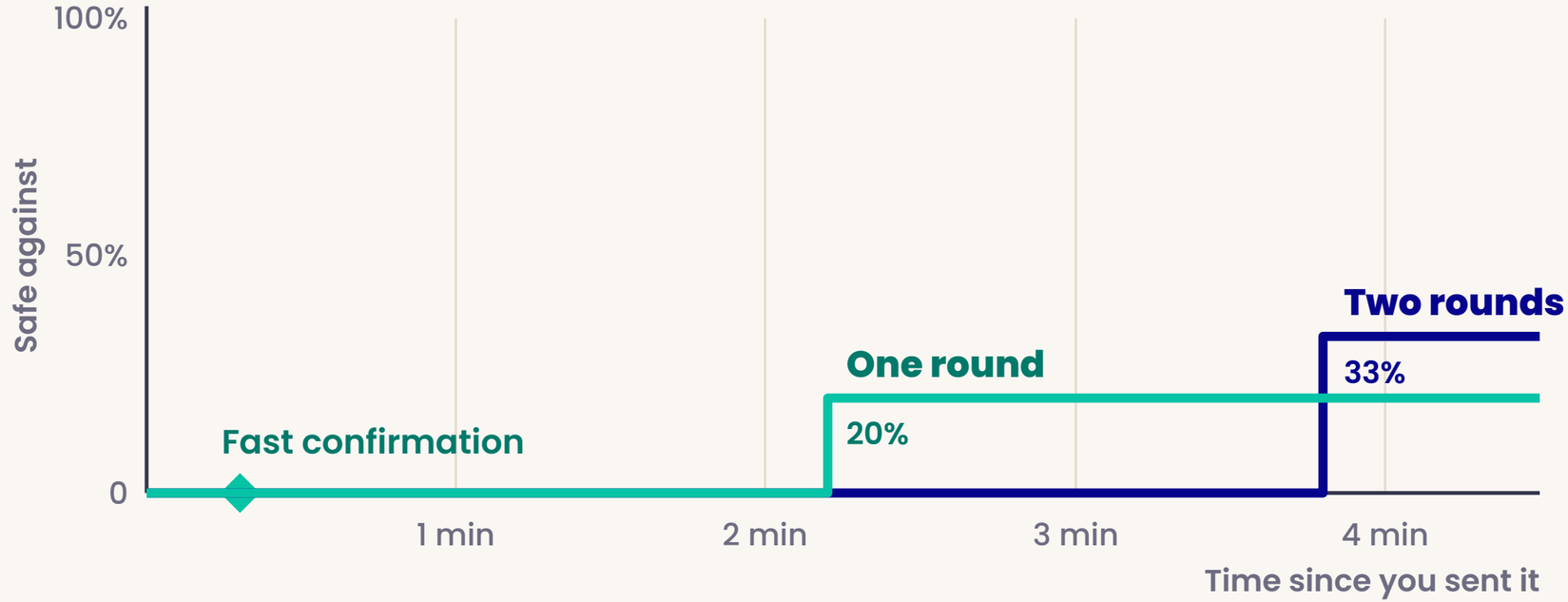
Faster, but weaker

final justified fast confirmed new checkpoint

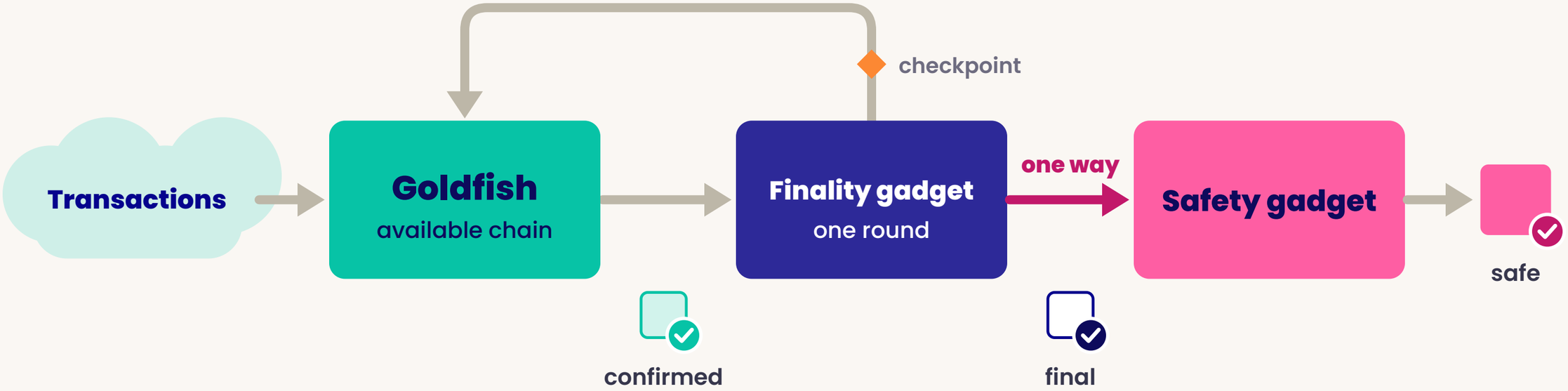
One-round finality
one round



2m 12s



New Result


checkpoint
Transactions
Goldfish
available chain
Finality gadget
one round
one way
Safety gadget
safe
confirmed
final

NEW PAPER

Safety gadget

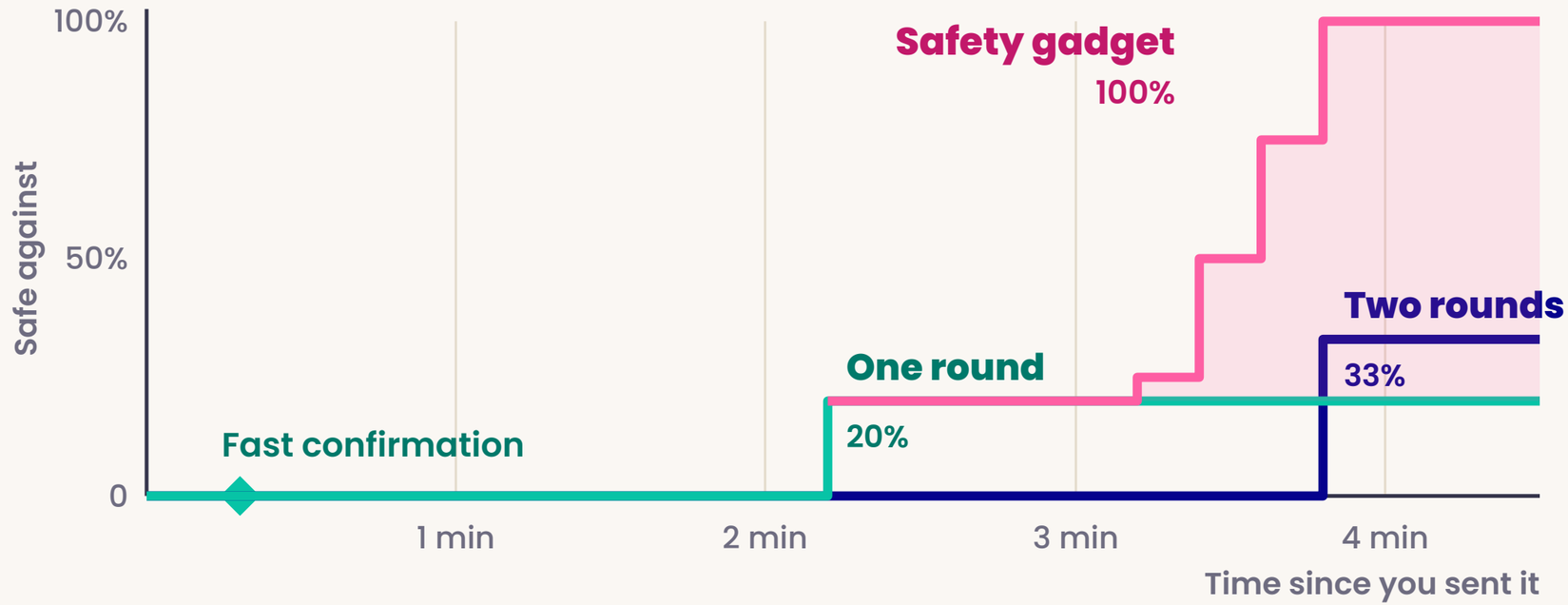
Short Paper: Flexible Consensus in Two and Three Rounds

Abstract. We apply OFLEX's checkpointing-style safety gadget on top of an off-the-shelf base consensus protocol like Minimmit with two-round good-case latency. This yields two interesting results. First, from a practical perspective, two-round protocols suffer from lower 20% resilience (compared to 33% for regular three-round protocols), which the gadget makes up for: it gives clients the option to trade off an extra round of latency for safety resilience up to 100%, or to remain at two rounds and 20% resilience. Second, from a theoretical perspective, the resulting flexible consensus protocol supports a substantial segment of the optimal safety–liveness tradeoff in three rounds, improving over the state-of-the-art four rounds of the generic OFLEX construction over a three-round base protocol. This is a first step towards the round complexity of optimal flexible consensus for safety–liveness resilience tradeoff, which remains an interesting open problem.

Keywords: Flexible consensus · Byzantine fault tolerance · Round complexity

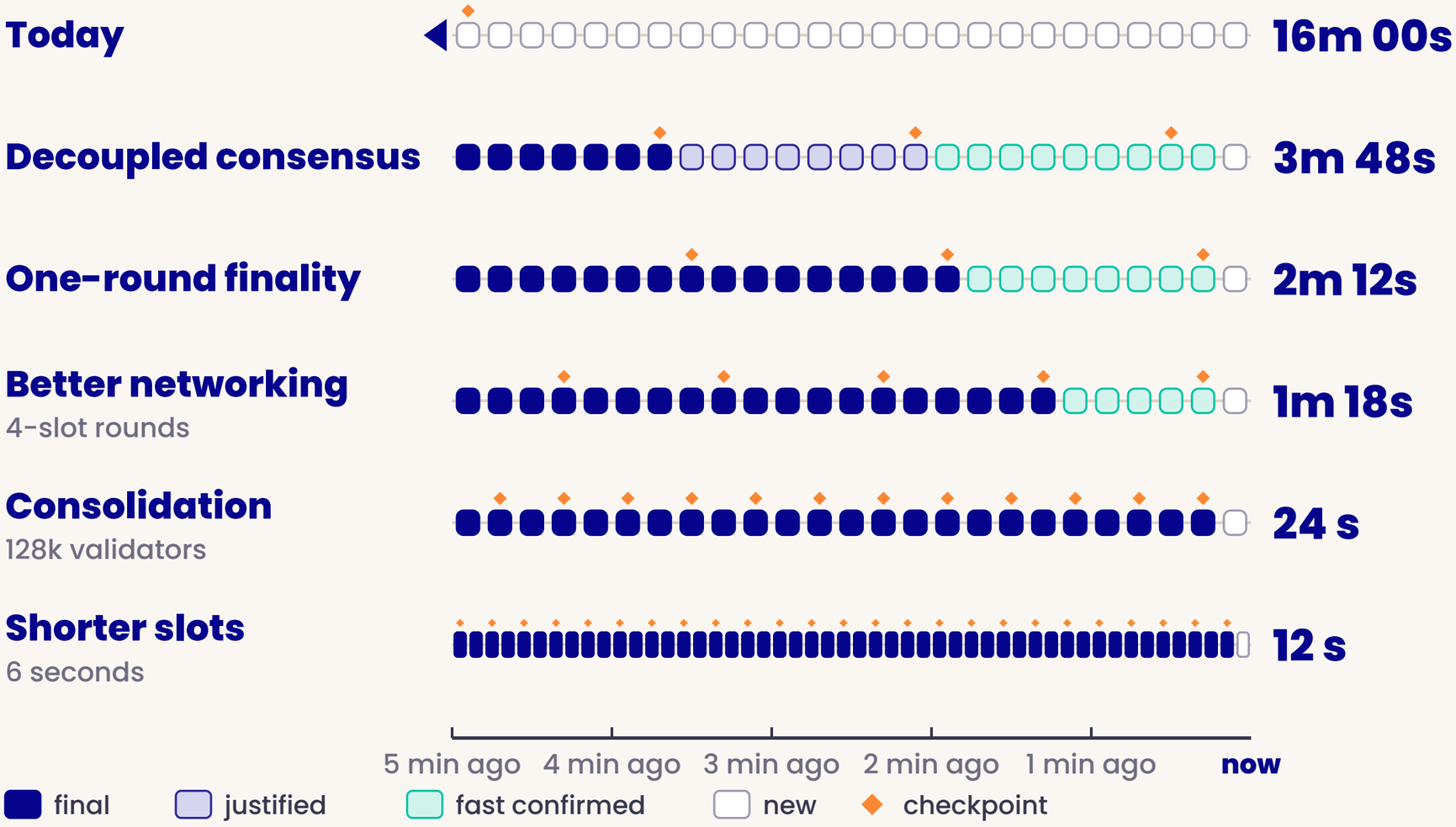
Optimal safety





STAGE 3 · SHORTER ROUNDS

It all multiplies



finality.watch

Turn the knobs yourself

Consolidate.

Fewer validators, shorter rounds.



consensus.ethereum.foundation/cambridge

This week at Devcon

Francesco D'Amato — Wed 10:30 · Lotus Stage 2

Fast Finality update

Barnabé Monnot — Wed 11:30 · Lotus Stage 3

Fast Ethereum

Julian Ma — Wed 17:10 · Lotus Stage 1

Fast Confirmation Rule

Roberto Saltini — Thu 15:00 · Workshop 1

Formal Verification of Consensus Protocols in Lean

Scan to add all four

to your Devcon schedule

The future is bright.



16 minutes → seconds

Rebuilding proof of stake, in stages.



Yann Vonlanthen

@yannvon

yann.vonlanthen@ethereum.org

finality.watch

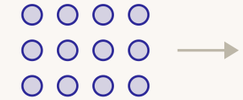
Cambridge talks

Devcon talks

Appendix

Not part of the five minutes. For Q&A.

What this unlocks



committee PQ signatures, as they are

Post-quantum readiness

Small committee, PQ signatures as they are



Fixing Gasper

Reorg-resilient chain, separate from finality



Fast finality, quick slots

At least 4× faster, off the critical path

Steps ahead

3 Nov 2026

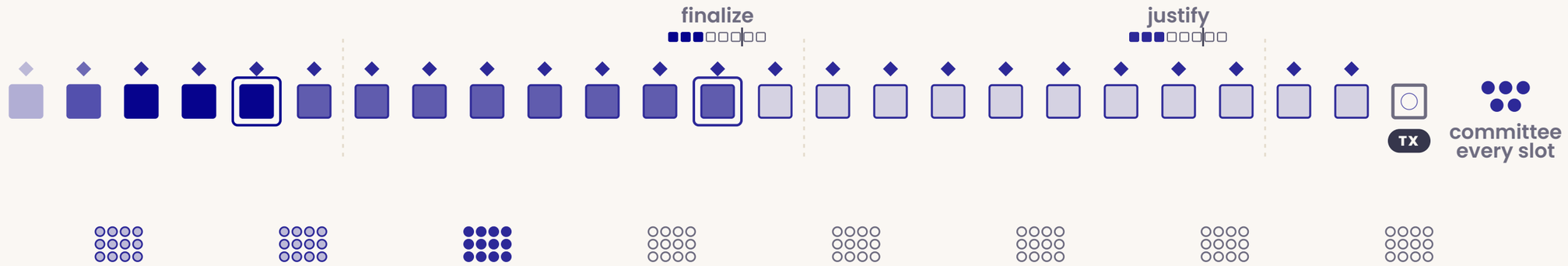
Mumbai

Devcon 8

We are here

Decoupled consensus, as it stands

Available chain · Goldfish with locks



Finality gadget · full validator set, one cohort per slot

One transaction, inclusion to finality

~3m 48s expected · today ~16 min

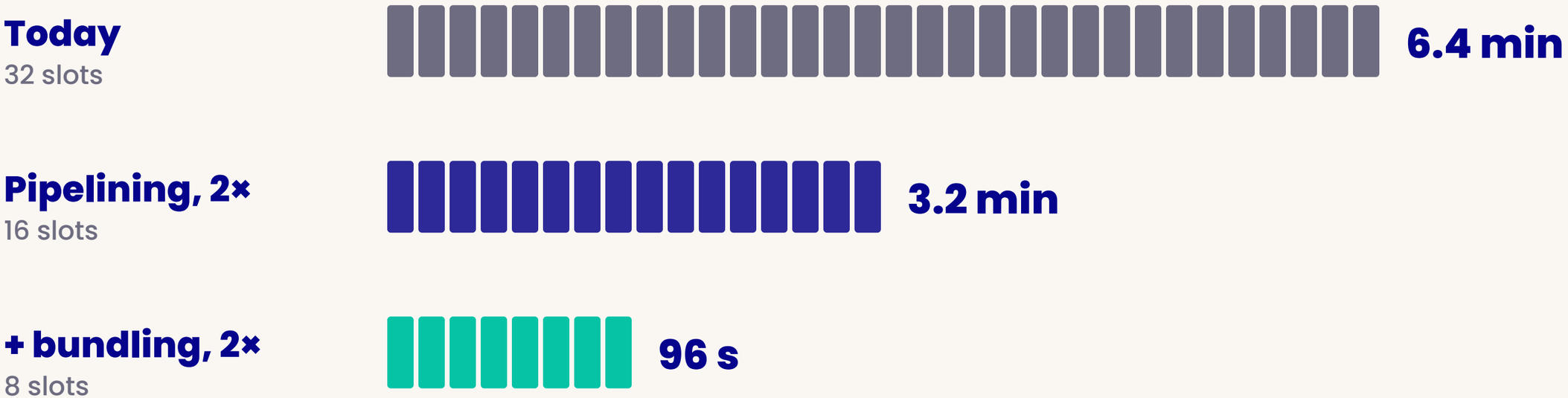
included

Inclusion	Confirmation	Wait for its round	Justification	Finalization
6 sec	12 sec	54 sec	1m 12s	1m 24s

NETWORKING

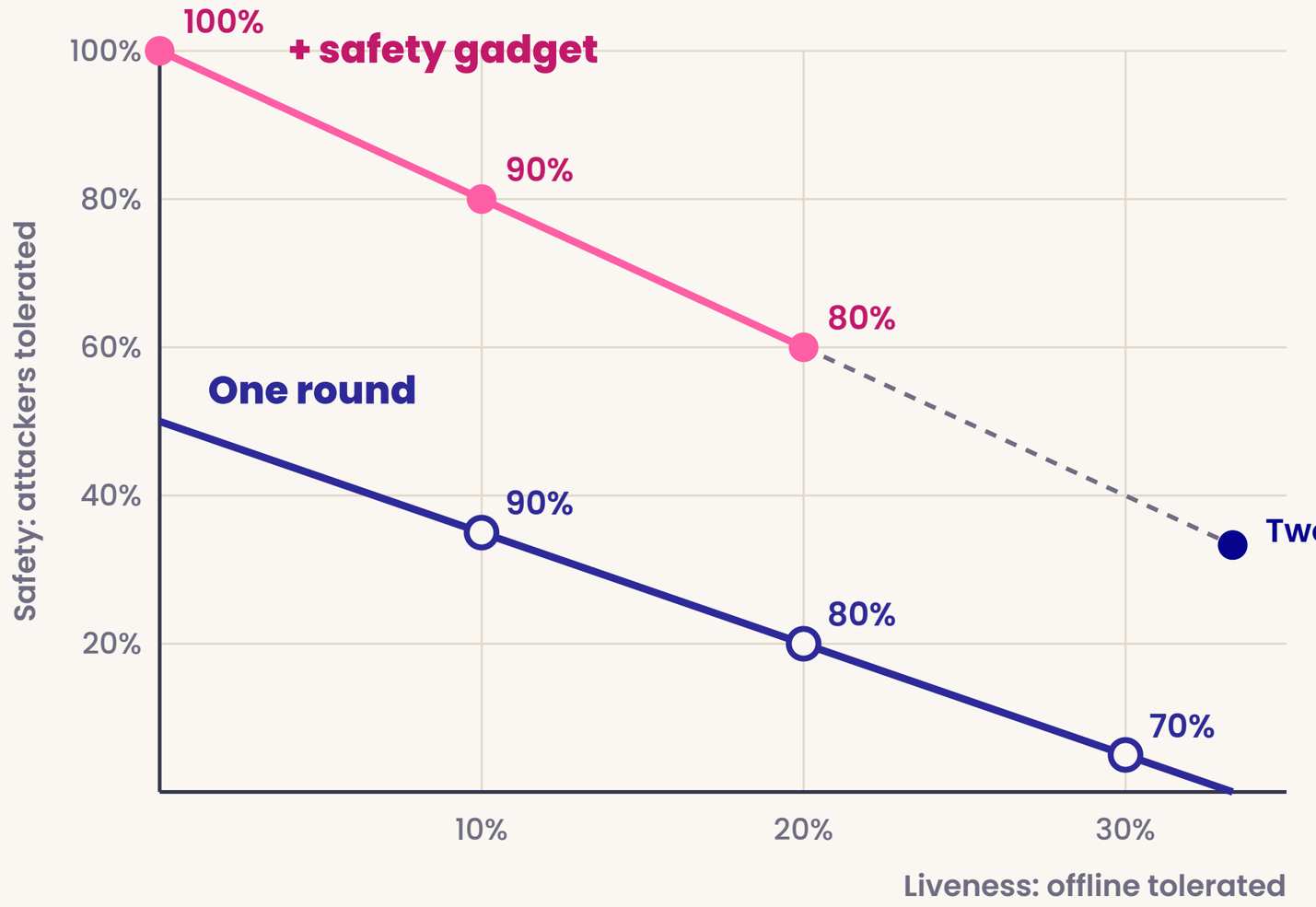
From 32 slots to 8

2× and 2× make the 4× faster finality we’re after. One box is one slot.



Pipelining: more committees per slot. Bundled attestations (EIP-8334): one copy of the data, many signatures.

One round: safety vs. liveness



Everyone

Finalize at 80% of votes

safe 20% · live 20%

Observers: you choose

Wait for 80%

safe 60% · live 20%

Wait for 90%

safe 80% · live 10%

Wait for 100%

safe 100% · live 0%

2t + 3p < n: Lamport ’02, Dutta et al. ’05 · gadget frontier: OFlex ’24 · labels are the share of votes needed